Singapore Partnership and Establishment Strategy

Oncology | Cell and gene therapy, primarily CAR-T — prepared for a Series A biotechnology company considering establishing a research presence in Singapore

Executive summary

Executive takeaway: research collaboration and technology-licensing discussions can begin before the company establishes its own Singapore entity. Applying directly to government grant schemes open to companies requires an eligible Singapore entity. For regulated clinical trials, a Singapore-registered local sponsor is required; this may be the company's own Singapore entity, or a suitable local organisation that agrees and is able to assume the applicable sponsor obligations.

The two tracks can run in parallel; they are not an either-or choice:

	Before establishing your own Singapore entity	After establishing your own Singapore entity
Research collaboration / technology licensing	Available through research agreements with local institutions and joint applications for academic grants	The company may also be eligible to act as lead applicant, subject to scheme-specific criteria
Clinical trials	The overseas entity cannot submit as the local sponsor – the submitting party must be a Singapore-registered entity. A suitable Singapore-registered CRO or institution may agree to assume the local-sponsor role and its associated obligations.	The company's Singapore entity may act as local sponsor, provided it can assume the applicable sponsor obligations
Government grant schemes for companies	Cannot apply directly	Some schemes become available, subject to local shareholding thresholds (see below)

Government-linked investment	Engagement can begin	Stated investment criteria emphasise locating key capabilities in Singapore
Manufacturing capacity	Engagement can begin, but scheduling priority is lower	The company's own Singapore entity may strengthen access and negotiating position at the stages with the fewest qualified providers

The recommended approach is to begin immediately with activities that do not require the company's own Singapore entity, while advancing incorporation in parallel. Incorporation does not, by itself, secure grant eligibility, sponsor readiness, investment or manufacturing capacity.

We assessed the parts of Singapore's cell and gene therapy ecosystem relevant to oncology across six dimensions and shortlisted 51 organisations for direct outreach. We also identified 4 potentially applicable grants, 5 EDB tax and investment incentives for which local shareholding is not stated as an eligibility criterion, and 2 relevant regulatory components, with 1 further component that applies conditionally.

Six ecosystem dimensions: direct outreach, routes via a public-sector partner, and items pending confirmation

Dimension	**Count**
Clinical research sites	4
Service and manufacturing	13
Grants and capital	15
Regulatory components	3
Laboratory space and siting	7
Industry partners	23

Annotations: +8 via public-sector partner | +3 pending | +4 pending

Four priority conclusions:

1. Plan manufacturing around multiple providers. Of the 7 critical stages, 4 have two or fewer providers in Singapore. Negotiate stage by stage, and treat those stages as critical-path

constraints when securing capacity and timelines.

2. Use a qualified Singapore local sponsor for regulated clinical trials. An overseas company cannot submit directly as the local sponsor. Either establish a Singapore entity able to assume the applicable sponsor obligations, or appoint a suitable Singapore-registered organisation by agreement.
3. Define outreach objectives and information boundaries before engagement. Of the 23 industry organisations identified, 14 operate in the same therapeutic area and modality, which creates both partnership opportunities and competitive sensitivities.
4. Evaluate how to reduce single-supplier exposure. For each of 4 critical stages, only one provider has currently been confirmed as capable of GMP delivery. Before finalising the operating model, compare selective in-house development, dual sourcing, long-term capacity agreements and contingency arrangements (see Section 3.2 and Section 5).

1. How we have framed the brief

The company faces three linked decisions: how to build partnerships in Singapore, how to establish a local presence, and which organisations to approach in what sequence. These decisions are interdependent: the outreach sequence depends on the chosen establishment route, which is in turn shaped by regulatory and funding requirements. The report is therefore structured as a sequenced establishment pathway, not as a directory of organisations.

The analysis is based on the company profile below. These are decision-relevant assumptions; if any of them changes, the conclusions may also change.

Organisation type	Company (not an academic institution)
Funding stage	Series A

Development stage	Phase 1 — the company has a clinical-stage candidate rather than a discovery-stage programme
Indication	Solid tumours
Therapeutic area	Oncology
Modality	Cell and gene therapy, primarily CAR-T
Headquarters	US
Singapore entity	Not yet incorporated

Two points about how we have read the brief:

- We have treated CAR-T as the primary modality, rather than the sole scope of the assessment. We assessed the full cell and gene therapy chain — binder design, vectors, cell collection, GMP manufacturing, release testing and cryogenic logistics — because the principal establishment bottlenecks often arise in shared upstream and downstream capabilities rather

than in the CAR-T modality itself.
- The absence of a Singapore entity is treated as a variable, not a fixed fact. The report states explicitly which conclusions change once an entity is established — that is the decision with the greatest influence on the available options.

2. The six dimensions we assessed

Establishing a cell and gene therapy programme means answering six different kinds of question at once. We assess them separately because each dimension has different constraints and a different degree of substitutability. Some clinical research sites may offer comparable alternatives; regulatory components, by contrast, are governed by fixed applicability criteria and are not interchangeable.

Dimension	What it answers	Character of the dimension
Clinical research sites	Which sites can conduct the trial	Comparable, subject to trial-specific requirements
Service and manufacturing network	Which providers cover each stage, from design through release	Assessed by stage rather than by company
Grants and capital	Which funding routes may be available, in what capacity the company may apply, and which investors may support a Singapore-based expansion	Eligibility depends on the applicant; some criteria may be met by changing which entity applies

3. Findings by dimension

3.1 Clinical research sites - 4 shortlisted sites

Singapore's oncology clinical-trial capacity is concentrated in a small number of public-sector institutions. Each shortlisted site has documented oncology research activity or relevant trial experience. The final column summarises publicly documented cell-therapy trial involvement, with inferred attribution and other uncertainties identified explicitly.

Institution	Type	Research focus and trial activity	Cell therapy trial record
KK Women's and Children's Hospital	Public specialty hospital	Obstetrics – High-Risk Pregnancy; Gynaecology – Gynaecological Cancers; Paediatrics – Childhood Cancers; Paediatrics – Genetic and Congenital Conditions; Active in clinical trials across various specialties.	See "KK Women's and Children's Hospital" below
National Cancer Centre Singapore	National specialty centre	Oncology – Immunotherapy; Oncology – Targeted Therapy; Cancer Genomics; Drug Discovery; More than 220 ongoing Phase 1 to IV local and international clinical trials.	See "National Cancer Centre Singapore" below

National University Hospital (NUH)	Public general hospital	Oncology – Cancer; Cardiology – Heart Diseases; Metabolic Diseases – Diabetes; Neuroscience – Neurology; Infectious Disease – Tropical Medicine; Actively conducts clinical trials across a wide range of medical specialties including Cancer, Cardiology, Metabolic Diseases, Neuroscience, and Infectious Diseases.	See "National University Hospital (NUH)" below
Singapore General Hospital (SGH)	Public general hospital	See "Singapore General Hospital (SGH)" below	See "Singapore General Hospital (SGH)" below

Supporting evidence by site follows.

KK Women's and Children's Hospital

Cell therapy trial record - NCT05429905 (dual anti-CD22/CD19 CAR-T, CART2219.1 - registry status UNKNOWN, last updated 2023-03-09); NCT05648019 (CD19 CAR-T, relapsed/refractory B-lineage leukaemia and lymphoma). Both are listed under "Transplant and Cell Therapies" on the research pages of the KKH paediatric haematology-oncology centre (principal investigators Dr Soh Shui Yen and Dr Michaela Seng).

National Cancer Centre Singapore

Cell therapy trial record - NCT04614103 (LN-145 TIL, metastatic NSCLC); NCT05727904 (lifileucel, LN-144, with pembrolizumab, melanoma - both belong to Iovance's autologous TIL product family; neither is a TCR therapy). These are adoptive cell therapies, but not CAR-T. Note also that within SingHealth, NCCS is a national specialty centre rather than a hospital: inpatient beds for early-phase work sit with SingHealth Investigational Medicine (SGH campus). NCT04912765 (neoantigen dendritic cell vaccine with nivolumab, hepatocellular carcinoma, Phase 2, RECRUITING).

National University Hospital (NUH)

Cell therapy trial record - NCT03282617 (dendritic cell vaccine CD137L-DC-EBV-VAX, nasopharyngeal carcinoma - status UNKNOWN, last updated 2017-09); NCT05038696 (multi-target CAR-T, ALaCART-B, B-ALL - as above; the registered site is a personal name and attribution is inferred; status UNKNOWN); NCT05043571 (anti-CD7 PEBL CAR-T, CARTALL, relapsed/refractory T-ALL - the registered site is a personal name, "Allen Yeoh Eng Juh"; attribution to NUH is our inference); NCT05302037 (allogeneic NKG2DL CAR-γ6T, CTM-N2D, developed by CytoMed, ANGELICA - an official NUHS release states an April 2023 start at NCIS, while the registered actual start date is 2024-11-19; both accounts are given, not merged); NCT06615479 (BMS-986393, arlo-cel, GPRC5D CAR-T, relapsed/refractory myeloma, Phase 3, recruiting); NCT07319676 (Epo-R-CD19 ± CD22 CAR-T, relapsed/refractory B-cell lymphoma, sponsored by NUH); NCT00717184 (autologous ex vivo activated NK cell infusion, pilot study, Phase 1, COMPLETED); NCT01974479 (redirected haploidientical natural killer cells, pilot study, Phase 1, SUSPENDED); NCT02030561 (NK cell infusion with trastuzumab, HER2-positive disease, Phase 1/2, UNKNOWN); NCT02123836 (natural killer cells in acute leukaemia and myelodysplasia, Phase 1, UNKNOWN); NCT02409576 (expanded, activated haploidientical natural killer cells, pilot study, Phase 1/2, UNKNOWN); NCT02507154 (reactivating NK cells in refractory head and neck cancer, Phase 1/2, UNKNOWN); NCT03242603 (combination immunotherapy in neuroblastoma, Phase 1/2, UNKNOWN).

Singapore General Hospital (SGH)

> Research focus and trial activity - Cancer - SMART Early Detection to Long-term Cancer Survival; Cardiovascular Emergencies - SMART Monitoring and Early Warning for Cardiovascular Emergencies; Diabetes and Metabolic Disorders - SMART Data Analytics and Innovative Technologies in Metabolic Diseases Monitoring; Infectious Diseases - SMART Precision Medicine for Priority Infectious Diseases Threats; Immunology and Transplant - Autoimmunity, Severe Allergy, Transplantation; Ageing and Population Health - SMART Care Models, Innovations and Risk Stratification for Graceful Ageing; Clinical Trials and Research Centre (CTRC), established in 1999, serves as SGH's clinical research hub; conducts trials across phases and therapeutic areas.
>
> Cell therapy trial record - NCT05648019 (CD19 CAR-T, relapsed/refractory B-lineage leukaemia and lymphoma); NCT07247201 (allogeneic expanded NK cells, AlloNK1, stage III-IV nasopharyngeal carcinoma - not yet recruiting). This is also the campus currently treating patients with commercially available Yescarta. NCT00394381 (autologous cytokine-induced

killer (CIK) cell adoptive immunotherapy, Phase 1/2, COMPLETED); NCT00460694 (allogeneic cytokine-induced killer (CIK) cell immunotherapy, relapsed disease, Phase 1/2, COMPLETED); NCT00815321 (Autologous Cytokine Induced Killer Cells (CIK) for Chronic, Phase 2, COMPLETED).

3.2 Service and manufacturing network - 13 organisations across 7 stages

This dimension should be assessed by process stage rather than by company. No single organisation covers the full CAR-T process end to end; a single-CDMO strategy would therefore leave at least one stage uncovered. The stages below are presented in process sequence.

Provider depth by supply-chain stage - 4 stages have two or fewer

Stages and counts:
- Research & Design: 1 (limited negotiating alternatives)
- Cell Therapy QC / Release Testing: 1 (limited negotiating alternatives)
- Clinical Manufacturing Process: 2 (limited negotiating alternatives)
- Cell & Gene Therapy Logistics: 2 (limited negotiating alternatives)
- Viral Vector Manufacturing: 3
- Cell Processing / Apheresis: 3
- Clinical Trial Execution – Phase 1/2: 6

Providers by stage are listed below. The notes after each name give the organisation's capability tags for that stage and its GMP status.

Research & Design (binder discovery, construct design, plasmid preparation) (1 identified provider; no GMP-confirmed provider)

- Esco Aster - Plasmid DNA Manufacturing, Cell Line Development, Cell Processing / Apheresis; GMP: self-declared cGMP; no HSA certificate located

Viral Vector Manufacturing (vector production) (3 identified providers; only 1 GMP-confirmed provider)

- CellVec - Viral Vector Manufacturing, BSL-2 Facility, Cell & Gene Therapy; GMP: HSA certified
- Esco Aster - Viral Vector Manufacturing, Cell Line Development, Cell Processing / Apheresis; GMP: self-declared cGMP; no HSA certificate located
- SGVector - Viral Vector Manufacturing, Bioanalytical Services, Cleanroom Fill & Finish; GMP: GMP certification could not be confirmed

Cell Processing / Apheresis (cell collection and processing) (3 identified providers; only 1 GMP-confirmed provider)

- Advanced Cell Therapy and Research Institute, Singapore (ACTRIS) - Cell Processing / Apheresis, Cell & Gene Therapy Manufacturing, Cell Therapy QC / Release Testing; GMP: HSA licence and GMP certificate (Feb 2025)
- Esco Aster - Cell Processing / Apheresis, Cell Line Development, Cleanroom Fill & Finish; GMP: self-declared cGMP; no HSA certificate located
- Lonza Biologics Tuas - Cell Processing / Apheresis, Allogeneic Cell Therapy, Cell & Gene Therapy Manufacturing

Clinical Manufacturing Process (clinical-grade manufacturing) (2 identified providers; only 1 GMP-confirmed provider)

- Advanced Cell Therapy and Research Institute, Singapore (ACTRIS) - Cell & Gene Therapy Manufacturing, Cell Processing / Apheresis, Cell Therapy QC / Release Testing; GMP: HSA licence and GMP certificate (Feb 2025)
- Lonza Biologics Tuas - Cell & Gene Therapy Manufacturing, Allogeneic Cell Therapy, Cell Processing / Apheresis

Cell Therapy QC / Release Testing (release testing) (1 identified provider; only 1 GMP-confirmed provider)

- Advanced Cell Therapy and Research Institute, Singapore (ACTRIS) - Cell Therapy QC / Release Testing, Cell & Gene Therapy Manufacturing, Cell Processing / Apheresis; GMP: HSA licence and GMP certificate (Feb 2025)

Cell & Gene Therapy Logistics (cryogenic logistics) (2 identified providers; losing one leaves a single provider)

- Cryoport - Cell & Gene Therapy Logistics, Cryogenic Storage
- World Courier Singapore - Cell & Gene Therapy Logistics

Clinical Trial Execution - Phase 1/2 (trial execution) (6 identified providers)

- CSI Medical Research - Phase 1/2, Clinical Trial Management, Clinical Trials
- Celero Global - Phase 1/2, Biostatistics & Data Science, Clinical Trial Management
- ClinActis - Phase 1/2, Biostatistics & Data Science, Clinical Trial Management
- Emerald Clinical Trials - Phase 1/2, Biostatistics & Data Science, Clinical Trial Management
- Expecto Health Science - Phase 1/2, Biostatistics & Data Science, Clinical Trial Management
- Parexel - Phase 1/2, Biostatistics & Data Science, Clinical Trial Management

One distinction matters more than the headline count: the number of providers that can be approached and the number that can deliver under GMP are two different figures. The table below gives both for each stage. The GMP column counts only organisations whose certificate scope covers that stage (see the notes against each provider above).

Stage	Identified providers	GMP-confirmed providers	Named GMP-confirmed provider(s)
Research & Design	1	0	—
Viral Vector Manufacturing	3	1	CellVec
Cell Processing / Apheresis	3	1	Advanced Cell Therapy and Research Institute, Singapore (ACTRIS)
Clinical Manufacturing Process	2	1	Advanced Cell Therapy and Research Institute, Singapore (ACTRIS)
Cell Therapy QC / Release Testing	1	1	Advanced Cell Therapy and Research Institute, Singapore (ACTRIS)
Cell & Gene Therapy Logistics	2	Not applicable	—
Clinical Trial Execution – Phase 1/2	6	Not applicable	—

3.3 Grants and capital - 4 the company may apply for directly, 8 public-institution-led schemes reviewed (company participation confirmed for some and pending individual verification for others), 5 EDB incentives with no stated shareholding criterion, and 6 government-linked investors

This section distinguishes four funding routes by source, applicant role and economic form: direct company applications, public-institution-led schemes, EDB fiscal incentives and equity investment. Groups A and B cover grant schemes, group C covers fiscal incentives, and group D covers equity capital. Establishment choices affect each route differently: shareholding structure determines eligibility for group A, the scale of local activity determines the strength of groups C and D, and group B requires a public-sector research partner.

Within groups A to C, the difference is not the amount but the capacity in which the company applies and which eligibility criteria apply. Group D is dilutive and is assessed on different criteria again.

A. Schemes the company may apply for directly:

Scheme	Administering agency	What it supports	Preconditions
Startup SG Tech (Proof-of-Concept / Proof-of-Value)	Enterprise Singapore	Funding for startups to develop proprietary technology solutions with commercial potential. Proof-of-Concept (PoC): up to S$400,000. Proof-of-Value (PoV): up to S$800,000. Amounts raised effective 2 January 2025. EnterpriseSG may take equity equivalent to grant amount.	The applicant must be incorporated in Singapore; the official page does not state the shareholding threshold, so obtain written confirmation from Enterprise Singapore

Enterprise Development Grant (EDG)	Enterprise Singapore	Broad-based grant supporting Singapore companies in upgrading capabilities, innovation, and overseas expansion.	The applicant must be incorporated in Singapore; and at least 30% local shareholding, assessed by look-through to ultimate individual shareholders – a wholly US-owned Singapore subsidiary does not qualify
EDGE (Enterprise Development Growth and Expansion)	Enterprise Singapore	Unified grant combining EDG, PSG and MRA. Supports digitalisation, market expansion, and enterprise efficiency. Not yet open for applications.	The applicant must be incorporated in Singapore; eligibility criteria not yet published
Startup SG Equity	Enterprise Singapore	Equity co-investment scheme for deep-tech startups. Government invests alongside qualified private investors. Extended to growth-stage companies at Budget 2026.	The applicant must be incorporated in Singapore; the official page does not state the shareholding threshold, so obtain written confirmation from Enterprise Singapore

B. Schemes led by a public institution or hospital, with the company participating as an industry partner: 8. These are available, but in a different capacity. Each is listed below. Where the participation route is blank, we have not yet verified that entry individually and say so.

Scheme	Administering agency	Why the company cannot apply directly	Capacity in which it can participate

Industry Alignment Fund - Pre-Positioning (IAF-PP)	Agency for Science, Technology and Research (A*STAR)	The applicant must be: public research institutes and universities — companies are not eligible applicants	A company cannot be the applicant (the lead PI must hold at least 0.7 FTE at a Singapore public institution). Industry participates through "early industry involvement"; the published wording is that the work is "expected to lead to industry participation within 3-5 years", without defining a formal partner role, and no official page states a co-funding requirement. Note the window: A*STAR states that "Applications for RIE2025 HHP and MTC IAF-PP FI are now closed", and as at 2026-08-10 the RIE2030 successor arrangements had not been published. Confirm the opening date with A*STAR before participating.

Open Fund - Individual Research Grant (OF-IRG)	National Medical Research Council	The applicant must be: public research institutes and universities — companies are not eligible applicants	Not individually verified
Clinician-Scientist Individual Research Grant (CSIRG)	National Medical Research Council	The applicant must be: public research institutes and universities or public hospitals and clinical institutions — companies are not eligible applicants	Not individually verified
Clinician Scientist-New Investigator Grant (CS-NIG)	National Medical Research Council	The applicant must be: public research institutes and universities or public hospitals and clinical institutions — companies are not eligible applicants	Not individually verified
Competitive Research Programme (CRP)	National Research Foundation	The applicant must be: public research institutes and universities — companies are not eligible applicants	Not individually verified

Industry Alignment Fund - Industry Collaboration Projects (IAF-ICP)	Agency for Science, Technology and Research (A*STAR)	The applicant must be: public research institutes and universities or public hospitals and clinical institutions - companies are not eligible applicants	A company cannot be the applicant, but can take part as an industry partner. A*STAR states verbatim: "IAF-ICP is open to all Public Research Institutions in Singapore" and "Companies are not eligible, nor will they be considered as co-applicants" (repeated verbatim on openinnovationnetwork.gov.sg - two sources). The scheme is designed around a public institution leading with industry co-funding. This report does not state the co-funding ratio (leverage multiple and cash proportion) - it appears only in second-hand accounts, no A*STAR page confirms it, and it is a figure that feeds directly into a budget. Request it in writing from A*STAR (IAF_ICP@a-star.edu.sg) if needed.

C. EDB corporate tax and investment incentives: 5. Here the company is itself the applicant, and no local shareholding percentage is stated as a criterion. What is assessed is the scale of investment and the substance of activity located in Singapore. For a wholly US-owned subsidiary, this route is more realistic than the group A grants:

Incentive	Administering agency	What it supports	Nature of the eligibility gate
Development and Expansion Incentive (DEI) — Manufacturing	Economic Development Board	concessionary rate of 5%, 10% or 15% on qualifying income in excess of the base income	No stated eligibility restriction based on company funding stage
Intellectual Property Development Incentive (IDI)	Economic Development Board	concessionary rate of 5%, 10% or 15% on a percentage of qualifying IP income	No stated eligibility restriction based on company funding stage
Pioneer Certificate Incentive (PC)	Economic Development Board	corporate tax exemption on qualifying income	No stated eligibility restriction based on company funding stage
Refundable Investment Credit (RIC)	Economic Development Board	A credit of 10%, 30% or 50% of qualifying expenditure; any unused portion is paid out in cash within four years	No stated eligibility restriction based on company funding stage

Enterprise Innovation Scheme - 400% R&D deduction	Economic Development Board	A 400% deduction on the first S$400,000 of qualifying R&D expenditure; up to S$100,000 may be converted to cash at 20% (capped at S$20,000 per year)	No stated eligibility restriction based on company funding stage

D. Capital: government-linked investment and private funds - 6 organisations. The first three groups covered non-repayable, non-dilutive funding. For a Series A company, though, the material leverage in Singapore sits on the other side: government-linked investors engage because capability is being located here. This assesses the same thing as group C (EDB incentives): how much the company intends to place in Singapore.

Government-linked investors in Singapore:

Organisation	Government linkage	Stated stage focus	Verifiable investments in this field
EDBI Pte Ltd	EDB subsidiary (Singapore Economic Development Board)	Growth; Late Stage	Esco Lifesciences Group, ImmunoScape, Tessa Therapeutics (dissolved)
Temasek Holdings	Singapore Government (state investor)	Venture; Growth; Pre-IPO; Buyout	MediSix Therapeutics, Tessa Therapeutics (dissolved)
ClavystBio	Temasek (via CLA Real Estate Holdings, wholly-owned subsidiary)	Early Stage	Paratus Sciences

Heliconia Capital Management	Temasek (wholly-owned subsidiary)	Growth	Tessa Therapeutics (dissolved)
SEEDS Capital	Enterprise Singapore subsidiary	Seed; Series A	No citable record at present
SG Growth Capital	Singapore Government (merger of EDBI and SEEDS Capital, eff. 1 Apr 2025)	Seed; Series A; Growth; Late Stage	No citable record at present

Funding rounds into companies in this field that are recorded in the knowledge graph (16 rounds; the 8 most recent are shown):

Company	**Round size**	**Date**	**Investors (bold indicates lead)**	**Status**
Tikva Allocell Pte Ltd	US$8.0M	2026-07-22	Kantharos Capital	Active
CytoMed Therapeutics	US$0.5M	2025	iCapital Holdings (SG) Pte. Ltd.	Active

Paratus Sciences	US$100.0M	2023	ARCH Venture Partners, Alexandria Venture Investments, ClavystBio, EcoR1 Capital, Leaps by Bayer, Polaris Partners	Active
Paratus Sciences	US$40.0M	2023	Leaps by Bayer	Active
Juniper Biologics	US$10.0M	2023	UOB Venture Management	Active
ImmunoScape	US$10.0M	2023	ScaleReady	Active
CytoMed Therapeutics	US$10.0M	2023	INCE Capital	Active
Juniper Biologics	US$5.0M	2023	Artemis Health Ventures Pte. Ltd.	Active

3.4 Regulatory components

Regulatory component	Administering agency	Scope of application	Key precondition

Clinical Trial Authorisation (CTA)	Health Sciences Authority (HSA)	Small Molecule; Biologics; Cell and gene therapy	Authorises the clinical trial. Submission must come from a Singapore-registered entity; an overseas sponsor cannot submit directly.
CTGTP Regulatory Framework	Health Sciences Authority (HSA)	Cell and gene therapy	Determines the applicable product classification and the resulting clinical, manufacturing and registration requirements. It is not itself a submission route.
GMAC environmental risk assessment	Genetic Modification Advisory Committee (GMAC)	Cell and gene therapy	Assesses the environmental and biosafety risk where the trial product contains genetically modified organisms. HSA advises allowing about six months and starting it in parallel rather than in sequence.

Two complementary regulatory components are relevant. The CTGTP framework determines the product classification and the requirements that follow; the CTA route governs authorisation of the clinical trial. They are not alternative pathways. A further 1 component applies conditionally, as shown in the table above. The distinction is easily missed:

- The CTGTP framework determines which class the product falls into, not which trial route applies. Autologous genetically modified T cells fall within Class 2 CTGTP (more than minimally manipulated, non-homologous use). That classification determines the rules that follow, and the Class 2 registration required at market authorisation.
- The CTA is the trial authorisation. HSA applies the CTA route to therapeutic products not registered locally and to Class 2 CTGTP. The CTN (notification) route is open only to products already registered locally and used within the approved label. The company's investigational CAR-T candidate is not locally registered; the CTA, rather than the CTN, is therefore the relevant trial-authorisation route.
- A third route, the CTC (Clinical Trial Certificate), applies to Chinese proprietary medicines and health supplements and is not relevant to cell and gene therapy.

A free pre-submission advisory channel is available. HSA operates an Innovation Office (Therapeutic Products and CTGTP) offering early scientific and regulatory advice to Class 2 CTGTP developers, covering non-clinical studies, clinical protocol design, CMC and manufacturing requirements. The process is to send the application form and a briefing document to HSA_InnovationOffice@hsa.gov.sg; HSA responds within 20 working days indicating whether it will convene a meeting or reply in writing. There is no charge. Note that the 20 working days is the deadline for the reply on how the matter will be handled, not for the meeting itself.

3.5 Laboratory space and siting - 7 sites meet the BSL-2 minimum

The three categories serve different purposes and are not interchangeable. Shared laboratories may offer flexible access to fitted space; science parks determine siting but do not themselves provide laboratory capacity; dedicated facilities are generally linked to a specific operating or partnership model.

Shared laboratories and incubators

- ClavystBio Node 1 - BSL-2; Science Park 1, Singapore. No public eligibility restriction was identified in the materials reviewed. ClavystBio's own siting guide recommends Node 1 alongside NSG BioLabs, co11ab and LSI for external start-ups. Opening coverage did state that the 13,000 sq ft "accommodates four of its 11 portfolio companies" - that describes current occupancy, not an eligibility rule. We suggest confirming directly with ClavystBio before approaching.
- Co11ab - BSL-2; Novena Health City
- Life Science Incubator (LSI) - BSL-2; Elementum (one-north) & German Centre (Jurong), Singapore
- NSG BioLabs - BSL-2; Biopolis Road & Science Park 1, one-north, Singapore

Science parks (a siting-level option, not a laboratory that can be leased directly)

- Biopolis - BSL-2 (leasable areas) | BSL-3 restricted to A*STAR high-containment facilities, not available to lease; one-north, Buona Vista Singapore
- Singapore Science Park - BSL-2 (selected buildings); Kent Ridge / Science Park Drive
- Tuas Biomedical Park - BSL-2; GMP-compliant manufacturing facilities also present; Tuas, Western Singapore

A further 3 sites are pending confirmation (the figure shown against this dimension in Figure 1). They are potential siting options, but cannot yet be counted as suitable laboratory space: the biosafety level is not published, and confirmation from the operator is needed before they can enter a siting plan:

- A*StartCentral - A*STAR states verbatim: "A*SC also provides various facilities and equipment to members, including wet labs and prototyping space." Wet laboratories are therefore confirmed, but no biosafety level is published. Using it requires confirmation from A*STAR first.
- BLOCK71 - Incubation space run by NUS Enterprise and available to occupy, but public material describes only "Collaborative spaces" and three incubation programmes in software and sustainability, with no mention of wet laboratories or a biosafety level. This is "not found", not "confirmed absent" - using it requires confirmation from NUS Enterprise first.
- JTC LaunchPad - JTC describes the site as "56,000sqm of modular units" (bare, fitted and semi-fitted), with target industries including biomedical sciences. It is shell space rather than a fitted laboratory, and no biosafety level is published. Placing a laboratory here means fitting it out and certifying it yourself.

3.6 Industry partners - 23 organisations

This dimension has the largest number of organisations, and it combines two distinct selection criteria. "Who competes in the same indication" and "whose capability fills a gap in the chain" are different questions, and one list cannot answer both. Section 4 sets out each selection separately:

Selection	Count	What it answers
Competitive overlap (Section 4.1)	14	Its pipeline overlaps with the company's therapeutic area and modality
Co-development fit (Section 4.2)	12	Its capabilities address a gap in the company's CAR-T development or manufacturing chain

Less: organisations selected under both criteria	3	Each organisation counted once

4. Competitive landscape and partnership positioning

23 potential partners: 14 with competitive overlap, 12 complementary by capability (3 in both groups)

Competitive overlap is assessed by pipeline (Section 4.1); complementarity by which stage a capability sits at (Section 4.2)

This section applies two different selection criteria to the same set of organisations, so a small number appear in both subsections:

4.1 Competitive overlap is assessed by pipeline. Whether an organisation competes with the company depends on the product it is developing, not on its list of capabilities.
4.2 Co-development fit is assessed by which stage of the CAR-T process a capability sits at. That is a separate question from whether the organisation competes.

4.1 Competitive overlap (14 organisations)

Has its own oncology cell or gene therapy pipeline. Define the scope of a conversation before opening it: these organisations are potential partners and potential competitors at the same time.

Organisation	Corporate stage	Capability tags	Basis for the assessment
Lion TCR	Series B	Cell Therapy Clinical Delivery, mRNA platform, Bench to Bedside	9 oncology cell or gene therapy programmes of its own; the most advanced is LioCys-M004 (LioCys-M) (Phase 2, 9 company-sponsored trials)

UTC Therapeutics Inc.	Series A	Allogeneic Cell Therapy, Phase 1/2, Gene Therapy Clinical Delivery	9 oncology cell or gene therapy programmes of its own; the most advanced is UCMYM802 (Phase 1, 16 company-sponsored trials)
Biosyngen	Growth	T-Cell Receptor Therapy, Cell Therapy Manufacturing, M-CEL	6 oncology cell or gene therapy programmes of its own; the most advanced is BRG01 (Phase 2, 8 company-sponsored trials)
Elpis Biopharmaceuticals	Series A	CAR-T Cell Therapy Clinical Delivery, Enzyme Engineering, Immuno-modulator	4 oncology cell or gene therapy programmes of its own; the most advanced is EPC-002 (Phase 1)
Neocytogen Therapeutics	Seed	CAR-T Cell Therapy Clinical Delivery, Cell Therapy Clinical Delivery, Oncolytic Virus Therapy	4 oncology cell or gene therapy programmes of its own; the most advanced is NCG101 (Preclinical)

CytoMed Therapeutics	Public	Allogeneic Cell Therapy, Allogeneic Cell Therapy Trials, CAR-T Cell Therapy Clinical Delivery	3 oncology cell or gene therapy programmes of its own; the most advanced is CTM-N2D (Phase 1, 3 company-sponsored trials)
Zeno Therapeutics	Series A	CAR-T Cell Therapy Clinical Delivery, Phase 1/2	3 oncology cell or gene therapy programmes of its own; the most advanced is GP350 CAR-T (Phase 2, 3 company-sponsored trials)
MediSix Therapeutics	Growth	Gene Editing (CRISPR/Cas), Immune Engineering, Bench to Bedside	3 oncology cell or gene therapy programmes of its own; the most advanced is PCART7 (Phase 1/2)
ImmunoScape	Series B	Cell Therapy Clinical Delivery, T-Cell Receptor Therapy, Computational Biology & Bioinformatics	3 oncology cell or gene therapy programmes of its own; the most advanced is IS-001 (Phase 3, 1 company-sponsored trial)

Tikva Allocell Pte Ltd	Series A	Allogeneic EBVST, CAR-T Cell Therapy Clinical Delivery, Cell & Gene Therapy	3 oncology cell or gene therapy programmes of its own; the most advanced is TAVST01 (IND-Enabling)

4.2 Co-development fit: by the process stage a capability addresses

The previous group was selected by what each organisation is developing - that answers the competition question. Partnership is a different question: which step of the process its capability addresses. This group therefore uses a different criterion, grouping by capability area rather than by product.

Two points on how to read the tables below:

- Entries marked competitive overlap already appeared in the previous group. They are not excluded, but the conversation differs: the subject is the specific platform technology (licensing, an evaluation agreement), not joint development.
- The third column records the verified scope of application. Under one capability tag, "currently applied to cell therapy" and "currently applied to cultivated meat" are not the same thing, and the tag does not distinguish them. This column does.

Binder discovery and protein engineering (4 organisations) - The target-recognition domain of a CAR is typically an antibody-derived fragment. This step determines how precisely the target is selected, and it is the most upstream and most easily overlooked part of the chain.

Organisation	Specific capability	Verified scope of application

DotBio	Antibody Discovery & Engineering	DotBody™ is a fully human VH single-domain antibody (not an scFv or VHH), with an intracellular variant iDotBody™. Its site lists four engagement models, including "License DotBody technology to accelerate your pipeline" and "Joint development of bi- or tri-specific assets against your targets" - the clearest stated route to partnership of any organisation here. CAR-T application is not mentioned on the site.
Elpis Biopharmaceuticals Δ competitive overlap	Protein Engineering	An mRNA display discovery engine plus mSCAFold™ cytokine engineering. It also appears in Section 4.1 as a competitive overlap - the subject of a conversation is platform licensing, not joint development.
Hummingbird Bioscience	Antibody Discovery & Engineering	Strength lies in target-epitope selection and ADC linker-payload work. It has its own clinical pipeline (HMBD-001 and HMBD-501 have reached Phase 1) and a licensing precedent (HMBD-002 was licensed to Percheron in June 2025). The subject of a conversation is binder licensing, not discovery as a service.

OpenProtein.AI	Protein Engineering	A protein language model and protein-design cloud platform, directly applicable to affinity and developability optimisation of scFv and VHH binders. A tools provider with no pipeline of its own.

Antigen and target discovery (1 organisation) - A central challenge in solid tumours is identifying a sufficiently selective surface antigen. This step comes before the CAR itself: if the target is wrong, every step after it is wasted.

Organisation	Specific capability	Verified scope of application
Otter Bio	Biomarker Discovery	Holds both TCR therapy and biomarker discovery capability. It is the only organisation in this group developing T-cell receptor therapies itself, so a discussion of targets may reach a competitive boundary.

Gene editing and delivery (5 organisations) - Knock-in and knock-out determine allogeneic potential and tolerability; non-viral delivery is the route around the lentiviral capacity bottleneck.

Organisation	Specific capability	Verified scope of application

AGEM Bio ∆ competitive overlap	Non-viral genetic delivery platform	A non-viral delivery platform applied to engineering mesenchymal stem cells ("MSC 2.0"). The delivery technology itself is relevant to T-cell engineering, but its own application is not in T cells.
Avecris	Non-Viral Delivery Technology	SPRING DNA™ is a programmable gene-medicine platform: non-viral, non-integrating, redosable, with flexible payload capacity. No externally offered process-development or analytical service is stated on its site — the subject is the platform, not a service.
LambdaGen Therapeutics ∆ competitive overlap	Targeted Genome Insertion	LIGIT (λ-integrate site-specific recombination) inserts fragments over 10 kb at safe-harbour sites and is available for licensing; its own pipeline is iPSC-derived CAR-NK.
SiVEC Biotechnologies	Non-Viral Delivery Technology	"BactPac" uses engineered bacteria to deliver mRNA, siRNA, protein or CRISPR; its stated strength is tissues that viral vectors and LNPs do not reach. Ex vivo T-cell transduction is not a use case it claims.

ZettaLab	Gene Editing (CRISPR/Cas)	Provides cloud software for molecular cloning and CRISPR design (an ELN plus a sequence platform), not wet-laboratory gene-editing capability.

Process development and manufacturing science (2 organisations) - The transition from a laboratory process to a scalable manufacturing process. It determines cost of goods and batch-to-batch consistency directly.

Organisation	Specific capability	Verified scope of application
Bioprocessing Technology Institute (BTI, A*STAR)	Cell & Gene Therapy, Culture Media Development (serum-free / chemically defined)	The only public research institute in this group covering process development, serum-free media and cell lines together. The granularity matches (an institute rather than a university), and it states an industry collaboration route explicitly.
ChemT Biotechnology	AI-Driven Manufacturing	Describes itself as "building the AI layer for CMC"; CelMo™ addresses growth and yield for cells used in biologics production. No published case of application to a CAR-T process was located.

4.3 Materials, consumables and tools suppliers - 4 organisations (not included in the counts above)

The relationship here differs from the previous two groups: those are about doing something jointly, this one is about buying something to use. They are separated because the negotiation is a different negotiation - the first two concern scope and rights, this one concerns specification, batch and lead time.

Organisation	What it supplies	Which step it serves
AuctuCel	AuctuPrime chemically defined cell culture medium; AI-driven bioprocess optimisation tools	Cell Culture Media Development, Cell Processing / Apheresis
Cellivate Technologies	Nanotopographic cell culture surfaces (coverslips, dishes, microcarriers) and Booster, a xeno-free serum repla	Culture Media Development (serum-free / chemically defined)
Cytiva	Equipment and consumables for cell and gene therapy manufacturing (bioreactors, filtration, aseptic fill-finis	Cell Processing / Apheresis
InnoCellular	Stem cell culture media (ready-to-use and basal formulations, including the serum-free, xeno-free MesenPlify™	Cell Culture Media Development, Cell Processing / Apheresis, Culture Media Development (serum-free / chemically defined)

5. Bringing the findings together

5.1 Three workstreams, one shared enabler, three different gates

The preconditions in the regulatory, funding and supply-chain dimensions all point towards a Singapore presence, but each opens on a different condition:

A clinical trial application must be submitted by a Singapore-registered entity. Non-dilutive funding has two routes: local incorporation, or collaboration with a principal investigator at a public institution (8 schemes take the second route). At the manufacturing stages with fewest providers, scheduling access usually depends on a local relationship.

A Singapore presence supports all three, but does not by itself resolve any of them. The three gates are different: an eligible local sponsor for the trial application; an eligible applicant for funding, which turns on incorporation and shareholding or on a public-institution partner; and a qualified, reserved provider at each thin manufacturing stage. A CRO acting as local sponsor resolves the first and neither of the other two. That is why this report treats them as three workstreams to start together rather than one decision to take first.

5.2 The number of providers that can be approached, and the number that can deliver clinical batches

The value of the table in Section 3.2 lies in the difference between its two columns. At 3 stages more than one provider can be approached (Viral Vector Manufacturing, Cell Processing / Apheresis, Clinical Manufacturing Process), while only one at each is confirmed to deliver under GMP. A further 2 stages have only one identified provider (Research & Design, Cell Therapy QC / Release Testing). The additional providers in the first group remain useful: they can support process development, research-grade batches and preparatory work before the company holds its own licence. They are not suitable for clinical batch delivery.

What follows from this is a scope for building in-house, not a conclusion that in-house is forced by unavailability:

- Release testing and process development are the first candidates for in-house capability. The headcount and equipment involved are contained, and both affect release speed and batch-to-batch consistency directly — the two things most exposed to an external provider's schedule.
- Vector production is not recommended for in-house build. It carries the largest capital requirement, and one third-party provider with certificate scope covering the stage does exist locally. The more practical approach is to secure the slot early and prepare a regional backup.
- Keep two options open for clinical-grade manufacturing. Two providers can be approached; for one of them, no extension of the licence to routine commercial supply was located. If commercial supply is the objective, this stage remains an open question.

These judgements rest on the public information gathered for this report. Actual scheduling and capacity still need confirming directly in first contact.

5.3 The three kinds of funding have different criteria, and therefore a different order of application

The term "non-dilutive funding" groups three things of different kinds. Their eligibility criteria differ, and so does the point at which each is worth applying for:

Category	Eligibility criterion	When to apply
Company grant schemes (Enterprise Singapore)	Local shareholding (EDG requires at least 30%, assessed by look-through to ultimate individual shareholders)	After the shareholding structure is settled; this criterion is difficult to adjust once incorporated
EDB tax and investment incentives (5)	Investment scale and substantive local activity; shareholding percentage is not a stated criterion	Once an investment plan exists; approach early, since these are forward-looking commitments
Equity funding	Requires an independent third-party lead investor first	Once a lead is committed, government-linked capital can follow on

For a wholly US-owned Singapore subsidiary the three differ markedly in reach: company grant schemes are constrained by the look-through shareholding test and will usually not be open; EDB incentives depend on the scale of local investment and are worth pursuing; equity funding is not constrained by local shareholding at all. Presenting the three as a single "schemes we can apply for" list invites the assumption that they can be pursued simultaneously - while the follow-on category is a consequence of a completed round, not a starting point.

5.4 With the regulatory position settled, the criteria for selecting a CRO and sites change accordingly

Three previously open questions now have answers. An overseas sponsor cannot submit a CTA directly. At the vector stage, only one provider is confirmed to hold a GMP certificate covering that step. Company grant schemes require a Singapore-incorporated company, and academic schemes do not accept a company as applicant. Taken separately these are three facts; taken together they bear on one thing - the criteria by which partners are selected.

Take the CTA point. It is usually treated as a filing detail, but what it actually changes is the selection criterion: when selecting a CRO, confirm whether it is willing and able to take the local-sponsor role. CROs differ in whether they are willing and able to act as local sponsor; some international CROs conduct execution locally without acting as sponsor. We recommend confirming this during the initial discussion rather than at protocol design.

5.5 The two negotiation tracks can run in parallel

The supply-chain providers in Section 3 and the co-development counterparties in Section 4.2 do not overlap. That follows from how each was selected: the first by whether an organisation can carry out a process step for others, the second by whether its technical capability is complementary. The two relationships differ in kind.

The practical consequence: the two negotiation tracks can run in parallel, with no required ordering between them, and need not be run by the same people. The manufacturing track concerns qualifications, scheduling and second sources; the technical track concerns evaluation agreements and licensing terms, and usually has a longer lead time. Run by one team, the technical track tends to slip under the manufacturing track's time pressure.

6. Recommended sequence

The order below follows from dependencies: the answer at each step affects the choices at the next. Durations are estimates based on each step's own published timelines and on common practice rather than on the dataset, which holds no cycle-time fields. They indicate sequence and order of magnitude, and are not a commitment on any party's response time.

6.1 Two tracks, run in parallel

	Track A: what can proceed without an entity	Track B: the establishment route
Objective	Build research and clinical relationships first	Resolve eligibility to sponsor a trial and to apply for funding directly
What the track produces	Substantive contact with sites, CROs and technical partners	A Singapore-registered entity
Key precondition	None	The shareholding structure settled first (look-through assessment is very hard to adjust after incorporation)
Earliest start	Now	The shareholding analysis can start now; incorporation waits on its result

Who is involved	Medical/clinical and business development	Company secretary / legal, with finance

Research collaboration, early HSA advice and first contact with CROs and sites do not require an entity first. Deferring these until after incorporation typically costs three to six months.

6.2 Timeline (month 0 is the start)

The three core dependencies below apply generally; the order of everything else is determined by staffing. Where the product contains genetically modified organisms, the GMAC assessment is a further external dependency whose timing is not controlled by staffing and which should begin in parallel.

Precondition	What it constrains	Basis
Shareholding structure settled	Incorporation	The look-through local shareholding percentage is very hard to adjust after incorporation
Incorporation complete	Applications for company grant schemes and EDB incentives (contact is not restricted)	The applicant must be a Singapore-registered entity
Local sponsor confirmed and protocol finalised	CTA submission	The submitting party must be a Singapore-registered entity, and the protocol needs site agreement

Everything else can start together in month 0, for the reasons given:

Can start now	Why it need not wait
HSA Innovation Office briefing document	Free, and no entity required; a reply within 20 working days, so the earlier it is asked the earlier the regulatory position settles

First contact with CROs and clinical sites	As above. The first question is whether they can act as local sponsor – of the three dependencies, this is the one whose answer rests with someone else
Scheduling discussions for the 4 stages with a single GMP-confirmed provider	Scheduling discussions can start before an entity exists, and slots are allocated in order of approach. Progress here depends on the provider's production calendar
Assessment of co-development counterparties (the four groups in 4.2)	No dependency on the regulatory, incorporation or manufacturing tracks. Binder discovery and non-viral delivery have the longest lead times and are therefore started first
Enquiries and site visits for shared laboratory space (3.5)	No entity required. In Singapore, shared laboratory space typically takes under a month from enquiry to occupation, so it is not on the critical path – but it determines when the team can arrive and start work
Entity form and shareholding analysis, plus contact with EDB	The analysis can be done ahead of incorporation; it is the input to the shareholding dependency above
Contact with government-linked investors	No entity is needed to make contact (as the summary table sets out). Terms wait on a lead investor, so these are exploratory conversations, not term discussions

Combining the two tables, the practical schedule is as follows (durations are order-of-magnitude estimates, not commitments):

Period	Track A (partnerships)	Track B (establishment)	What this period unlocks

Months 0–1 in parallel	(1) Submit the HSA briefing document. (2) First contact with CROs and sites, asking about local sponsorship first. (3) Scheduling discussions for the 4 stages with a single GMP-confirmed provider. (4) Approach the two co-development groups with the longest lead times – binder discovery and non-viral delivery. (5) Enquiries and visits for shared laboratory space.	(6) Entity form and shareholding analysis, with EDB contact in the same period to set out the medium-term investment plan. (7) First contact with government-linked investors, exploratory rather than on terms.	Three waiting periods start together: the HSA reply, the site responses and the manufacturing slots
Months 1–3	Once the HSA reply arrives, discuss the protocol with sites; write second-source provisions into contracts at the two-provider stages (see 6.3); approach the remaining two co-development groups; sign and occupy laboratory space	Shareholding settled, then incorporation completed (at least one resident director is required)	The regulatory position converges; manufacturing moves from an open question to a schedulable one; the shareholding and incorporation dependencies clear; the team can move in

Months 3–5	Protocol finalised; local sponsor confirmed	Fundraising becomes substantive, starting with an independent lead investor; grant and incentive applications can be submitted once incorporation is complete	The local-sponsor dependency clears; the funding structure takes shape

6.3 Three things worth settling early

Manufacturing slots at the 4 stages where only one provider is confirmed to hold GMP certification covering the step (Viral Vector Manufacturing, Cell Processing / Apheresis, Clinical Manufacturing Process, Cell Therapy QC / Release Testing). This limits both negotiating room and substitutability. We suggest securing an indication of scheduling first and assessing a regional backup. At 1 of them (Cell Therapy QC / Release Testing) that provider is also the only one identified at all. At the remaining 3 (Viral Vector Manufacturing, Cell Processing / Apheresis, Clinical Manufacturing Process), other providers can be approached but none is GMP-confirmed for the step — a second-source clause on its own does not create a second source; the backup has to be identified and shown to be technically qualified.

The shareholding structure. It determines access to company grant schemes and is the hardest element to adjust after incorporation. We suggest establishing, before the incorporation documents are finalised, whether the 30% local shareholding threshold needs to be met and whether it can be.

The local sponsor. What it settles directly is the regulatory workstream: the submitting party must be a Singapore-registered entity. We suggest holding off on commercial commitments to any single organisation until it is settled.

6.4 Three workstreams to begin in parallel

Start the three workstreams together rather than waiting on any one of them: (1) confirm an eligible local sponsor and settle the HSA position; (2) settle the shareholding and funding strategy, or secure an eligible public-institution partner; (3) qualify and reserve capacity at the thin manufacturing stages, and assess a regional backup in parallel. Each is subject to a different condition, so none of them substitutes for another - and none requires an entity to be in place before the conversations can begin.

6.5 Basis for the durations quoted in this section

The durations in this section fall into three kinds. The table below states which kind each one is, because that determines how finely you can plan against it.

Duration	Where the number comes from
Deferring research collaboration, regulatory advice and CRO contact until after incorporation typically costs three to six months	Industry norm; a planning assumption rather than a measured figure, and not a commitment
In months 0–1, running each item in parallel saves roughly a month	An order-of-magnitude estimate from the individual step timings in the table above, not a measurement
In Singapore, shared laboratory space typically takes under a month from enquiry to occupation	Local practice, as advised by the project team (17 Aug 2026); an industry norm rather than a measured figure
Manufacturing slots are allocated in order of approach	Common practice; individual providers' scheduling rules were not verified organisation by organisation
The HSA Innovation Office replies within 20 working days on whether it will arrange a meeting or respond in writing	Published by the HSA Innovation Office, in its published service procedure; confirmed 18 August 2026

Durations that come from the collected data (such as the CTA review time) are quoted directly from that record in the tables above, with the officially published figure or range preserved. The table here lists only those that do not come from the collected data.

Appendix: organisations we can introduce you to

Every organisation below comes from the findings in the sections above. Name any of them and we will prepare and submit an introduction request through the relevant contact channel.

A total of 51 organisations (an organisation appearing in two sections is counted once), grouped by where it appears in the report:

Clinical research sites (Section 3.1) (4): KK Women's and Children's Hospital, National Cancer Centre Singapore, National University Hospital (NUH), Singapore General Hospital (SGH)

Service and manufacturing network (Section 3.2) (13): Advanced Cell Therapy and Research Institute, Singapore (ACTRIS), CSI Medical Research, Celero Global, CellVec, ClinActis, Cryoport, Emerald Clinical Trials, Esco Aster, Expecto Health Science, Lonza Biologics Tuas, Parexel, SGVector, World Courier Singapore

Laboratory space and siting (Section 3.5) (7): Biopolis, ClavystBio Node 1, Co11ab, Life Science Incubator (LSI), NSG BioLabs, Singapore Science Park, Tuas Biomedical Park

Competitive overlap (Section 4.1) (14): AGEM Bio, Biosyngen, CytoMed Therapeutics, Elpis Biopharmaceuticals, ImmVesica Therapeutics, ImmunoScape, LambdaGen Therapeutics, Lion TCR, MediSix Therapeutics, Neocytogen Therapeutics, SCG Cell Therapy, Tikva Allocell Pte Ltd, UTC Therapeutics Inc., Zeno Therapeutics

Co-development fit (Section 4.2) (12): AGEM Bio, Avecris, Bioprocessing Technology Institute (BTI, A*STAR), ChemT Biotechnology, DotBio, Elpis Biopharmaceuticals, Hummingbird Bioscience, LambdaGen Therapeutics, OpenProtein.AI, Otter Bio, SiVEC Biotechnologies, ZettaLab

Materials and tools suppliers (Section 4.3) (4): AuctuCel, Cellivate Technologies, Cytiva, InnoCellular